Snowdon Towers Electrical VDC Capstone

Project Brief

Project information	Value
Project code	ST-VDC-01
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Date	September 26, 2026
Status	Approved for capstone planning
Project type	Independent educational electrical VDC capstone

Purpose

To consolidate what I learned through Autodesk's Revit training, I am developing an independent electrical VDC capstone using the multidisciplinary models provided in Autodesk's Snowdon Towers sample project. I selected a limited project area and will develop it through modeling, coordination, documentation, quantity reporting, revision, and closeout. I chose the deliverables by reviewing common electrical VDC workflows and recurring responsibilities across BIM and VDC job listings. The purpose is to move beyond guided tutorials and demonstrate how I organize, apply, and communicate my developing skills in a realistic project environment.

Training foundation

This capstone builds on Autodesk's Learn Revit in 90 Minutes, Build Electrical Systems in Revit, and Revit for Electrical Design Professional Certification Prep courses.

Project context

Autodesk's Snowdon Towers sample project supplies coordinated architectural, structural, HVAC, plumbing, site and facade models. These models establish the building context. I will create a separate electrical working model and use the other disciplines as linked references. Autodesk's completed electrical model will remain untouched and may be consulted only as a learning reference; it will not be presented as my work.

Primary outcome

The finished project will demonstrate a controlled workflow from received multidisciplinary models through electrical modeling, spatial coordination, field-oriented documentation, prefabrication planning, quantity reporting, revision and record-model closeout.

Scope

The capstone will cover one focused coordination zone rather than the complete tower. The exact location will be selected after reviewing the source models. The target zone will contain an electrical-room opportunity, a primary corridor distribution path and a small adjoining occupied area. Only the levels needed to demonstrate a meaningful vertical transition will be included.

Electrical focus

The primary focus is power distribution, raceway coordination and prefabrication. A limited number of devices may be used as believable endpoints, but the project will not include a complete lighting design.

- Two panels, one transformer and one disconnect
- Approximately eight to twelve electrical devices or junction boxes
- Conduit runs with fittings, bends, offsets and elevation changes
- One organized overhead conduit rack
- Cable tray with representative branch routes
- Equipment, device and material schedules
- At least three documented coordination problems and resolutions
- One relevant electrical-family modification
- One simulated field change followed through the model and record documents

Limitations

Snowdon Towers is an independent educational project and is not intended for construction. Autodesk sample models and content provide the building context. The project is limited to one selected coordination zone and does not represent a complete building electrical design. Equipment requirements, electrical characteristics and design criteria are treated as provided assumptions for modeling purposes. The project does not include engineering calculations, code certification, permit documents or professional approval. Coordination decisions, field conditions and revisions are simulated and clearly identified as such.

Excluded work

- Complete tower electrical design
- Service sizing, load calculations, voltage-drop studies or short-circuit analysis
- Photometric calculations, complete lighting controls or emergency-lighting design
- Fire alarm, communications, security and other low-voltage systems
- Professional code certification, stamped drawings or construction authorization
- Claims of actual site verification, contractor meetings or field installation

Required deliverables

Model deliverables

Deliverable	Purpose
Electrical working model	Original Revit model containing the selected electrical scope
Record model	Final Revit version incorporating the simulated field change
Coordination export	NWC export for federated review and clash coordination
Source-model register	Record of received Autodesk models, filenames, versions and project roles

Drawing package

Sheet	Title and purpose
E0.01	Cover and index with scope, assumptions, symbols and sheet list
E1.01	Electrical coordination plan showing the selected zone and primary routing
E2.01	Enlarged electrical-room plan showing equipment, dimensions and connections
E2.02	Raceway plan and sections showing routes, elevations, bends and congested areas
PF1.01	Prefabrication build sheet with one rack assembly and bill of materials
IS1.01	Installation sheet showing where and how the selected assembly is installed

Supporting records

Record	Purpose
Equipment and device schedule	Verify naming, parameters and model-data consistency
Demonstration panel schedule	Show circuit relationships using clearly identified educational assumptions
MTO-01 material takeoff	Connect modeled quantities to purchasing and prefab planning
Clash log	Document initial conditions, decisions and resolved conditions
Issues and decisions log	Track assumptions, questions, sources and resolutions
FC-01 field change	Track one simulated change through model and drawing updates
Model QA checklist	Review names, dimensions, elevations, connections, data and exports

Completion criteria

The capstone is complete when a reviewer can understand the project without opening Revit and can trace each public claim to a model view, drawing, schedule, report or decision record.

- The selected zone and all included elements are clearly identified.
- Linked models align without unexplained displacement.
- Equipment, conduit and tray are consistently named and scheduled.
- Routing elevations, bends, transitions and intersections are visible in plan, section and 3D views.
- At least three coordination problems are documented before and after resolution.
- The drawing package communicates locations, dimensions, elevations and installation intent.
- The prefab assembly reconciles with its bill of materials and model quantities.
- The simulated field change is traceable through the issue log, model, drawings and record package.
- RVT, PDF, DWG, NWC and spreadsheet exports use consistent filenames and open successfully.

Documentation rule

Every work session must end with a saved or versioned working file, a tracker update, at least one useful screenshot or export, any relevant issue or decision entry, and a clearly written next action. Public materials will be curated from this evidence rather than recreated after the project is finished.

Portfolio presentation

Component	Plan
Case-study PDF	Seven concise pages covering purpose, inputs, modeled scope, coordination, field documents, prefab and closeout
Project webpage	One static page hosted free through GitHub Pages with selected evidence and the case-study download
Video walkthrough	A narrated 60-90 second screen recording showing the model, one clash resolution and final deliverables
Resume	A Selected Technical Project entry separate from professional employment
LinkedIn	Optional Projects entry and a restrained learning-oriented post after the work is ready

Approved planning decisions

- Use the existing Snowdon Towers building models as received context.
- Create an original electrical working model.
- Limit the project to one focused zone.
- Prioritize power distribution, raceway coordination and prefabrication.
- Label all assumptions and simulated conditions.
- Use a concise case study, free webpage and short walkthrough.

Next phase

Session S02 will acquire and inventory the Snowdon Towers source models. Session S03 will compare candidate project zones and select the exact area. Session S04 will establish file structure, naming, view and sheet standards, assumptions, routing rules, capture procedures and the Gate 0 approval.